CFD PROJECT REPORT

Valve Leakage Simulation

ENG103 — Fluid Mechanics

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A computational fluid dynamics analysis of internal flow and pressure loss through valve leakage paths.

SOFTWARE

SimScale | OnShape

COURSE

ENG103 — Fluid Mechanics | Narugopal Ghata

DATE

March 16, 2026

Background

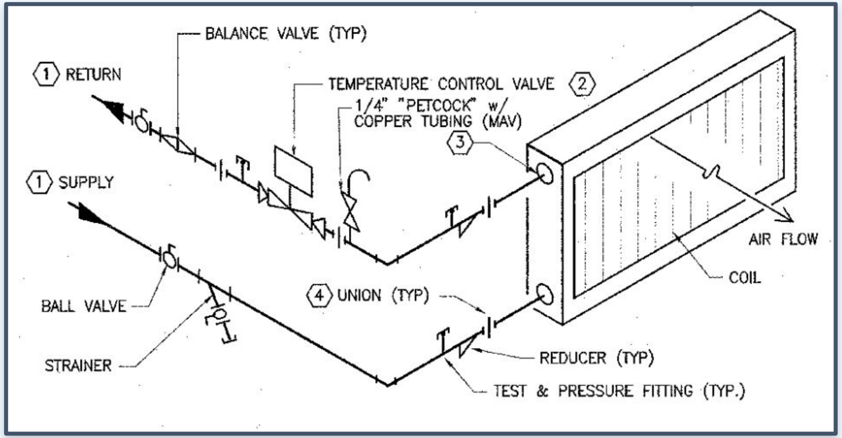

Fig. i — Giedt Reheat Diagram

Key Takeaways

Temperature Control valve is placed after exiting reheat coils and is typically a Globe Valve. This is actuated by a Siemens controller which is commanded when a zone calls for reheat.

Background

We come from a background of working with UC Davis Facilities Management: Energy Engineering, assisting their mission in saving energy across campus by analysis and diagnostics of HVAC equipment. One of the common problems we come across is reheat valve leakage which can occur when the valve wears out due to use over time.

(Note that this is not your typical type of leakage that may occur due to corroded pipe fittings, where fluid escapes the pipe)

This causes an unintentional discharge temperatures to rise after exiting the coils leading to wasted energy, warmer zone temperatures, and discomfort of occupants. In this simulation, we focused on the wasted energy, calculating exactly how much energy is lost from a gap of only 3mm.

1 Problem Description & Objectives

Problem Description

- Similute internal pressure-driven flow through a leaking 2-way globe valve seat in a hot water reheat system at Giedt Hall
- Flow Type: Internal, incompressible, turbulent single-phase liquid
- Real-world application: SkySpark fault detection identified zones with discharge air temps exceeding AHU supply air temp while reheat valves are commanded closed (due to valve leakage)

Project Objectives

- Calculate pressure drop across the valve seat leakage gap at 50-60 PSI system differential
- Determine leakage flow rate through a worn valve seat gap of 0.1-0.5mm
- Analyze velocity distribution and turbulence characteristics

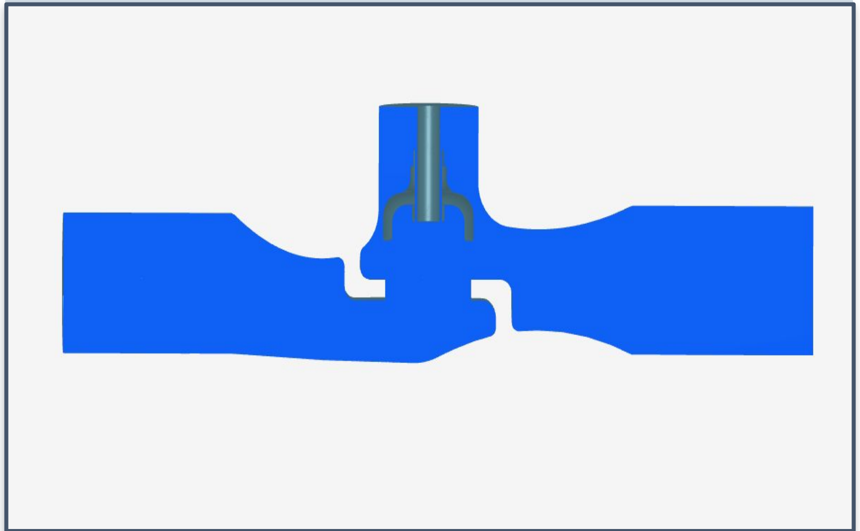

Fig. 1 — Geometry & Computational Domain

Key Dimensions

Valve gap: 0.3 mm | Inlet/Outlet Ø: 1” (25.4 mm)
Domain length: ~ 150 mm (roughly 6x pipe diameter)

2A

Simulation Setup — Mesh Generation

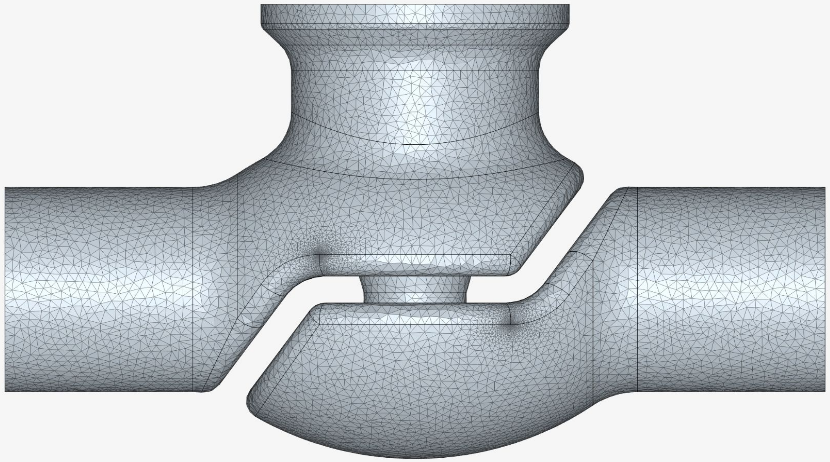

Fig. 2A — Computational Mesh

Mesh Type

Unstructured with inflation layers near walls

No. of Elements

313k Cells

No. of Nodes

80.5k Nodes

Mesh Quality

Skewness < 0.85°, Non-orthogonality < 70°

Near-Wall

Wall functions, 3 inflation layers, y^+ = 30-300

2B Simulation Setup — Mesh Generation (actually used)

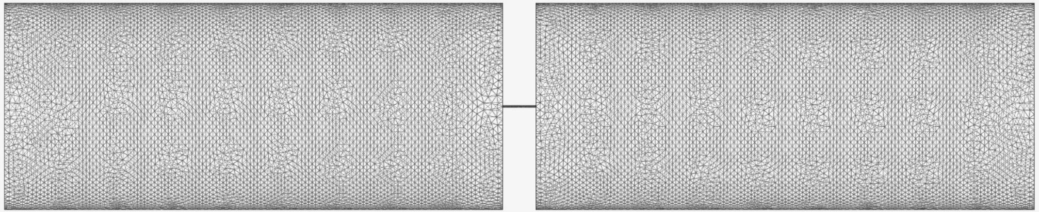

Fig. 2B — Computational Mesh

Mesh Type

Unstructured with inflation layers near walls

No. of Elements

488.7k Cells

No. of Nodes

160.85k Nodes

Mesh Quality

Skewness < 0.85°, Non-orthogonality < 70°

Near-Wall

Wall functions, 3 inflation layers, y^+ = 30-300

**NOTE: Due to the complexity of the previous model, computation was very difficult. In its place, we opted for a simplified valve leakage using a thin cylindrical pipe.*

Simulation Setup — Physics & Boundary Conditions

Physics Models

- Solver type: Pressure-based
- Turbulence model: k-ε realizable
- Energy equation: Off (isothermal assumption)
- Flow regime: Steady-state
- Fluid: Incompressible

Boundary Conditions

- Pipe Diameter: 25.4mm
- Inlet: P = 75 PSI (517 kPa)
- Outlet: P = 15-25 (103-172 kPa)
- Walls: No-slip, stationary
- Symmetry: Axis (2D axisymmetric)
- Valve seat: No-slip, stationary wall

Fluid Properties

- Fluid: Hot water
- Density: 971 kg/m^3
- Dynamic viscosity: 0.000352 Pa*s
- Temperature: 180°F/82°C
- Reynolds number of pipe: 9,440

  (> 4,000 -> turbulent flow)

Solver: SIMPLE | Discretization: 2nd order upwind | Convergence: residuals < 10^{-4} | Iterations: 500

4

Key Assumptions & Simplifications

01 **Steady-State**

Since we are not looking at variance in the flow or leakage, simulation is steady-state. We are looking at the cases where there is constant flow across the valve, and not when the valve is opening and closing, or changes in pressure during system startup and shutdown

02 **2D or 3D Domain**

Due to the geometry of the pipes and valves being axisymmetric, we will simulate using a 2D domain.

03 **Incompressible Flow**

Since we are working with HVAC operating conditions, the velocity of the chilled water is much smaller than the speed of sound in water. (Mach number well below 0.3)

04 **Other Simplifications**

We are assuming that the only water (no air) is flowing through the pipes and stays at ~180°F / 82°C throughout (no heat transfer).

5 Results & Analysis

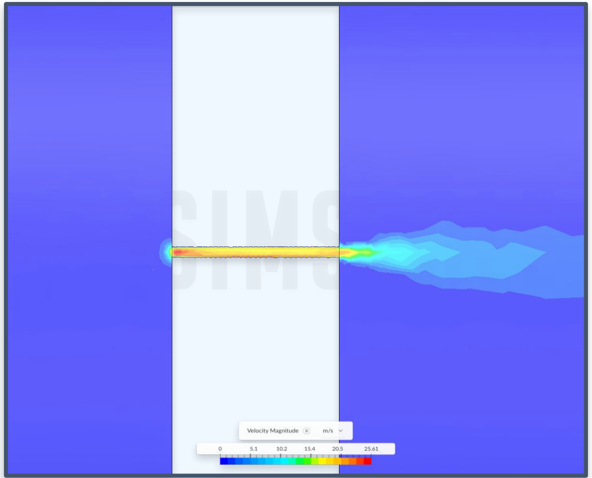

Fig. 3 — Velocity Contours

Fig. 4 — Pressure Distribution

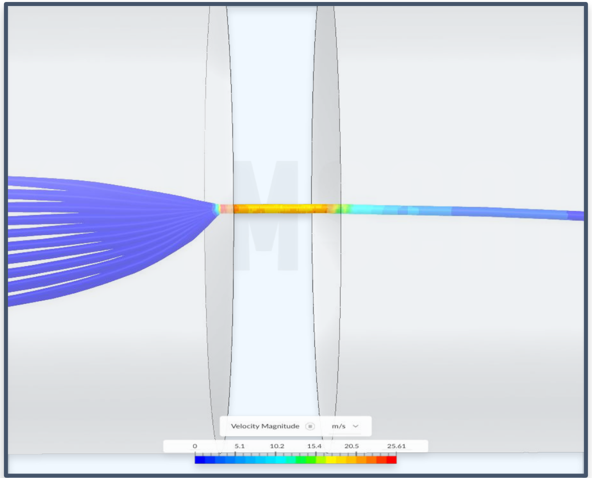

Fig. 5 — Streamlines

KEY FINDINGS

- Pressure drop: e.g. ΔP = 379,212 Pa (55 psi) across the leakage gap at V_{inlet} = 25.36 m/s
- Leakage flow rate: e.g. $\dot{m}$ = 0.00343 kg/s (≈ 3.36 GPH) through the 0.3mm gap
- High velocity jet confirmed at valve seat, turbulent wake and recirculation observed downstream of constriction

6 Results — Additional Visualizations

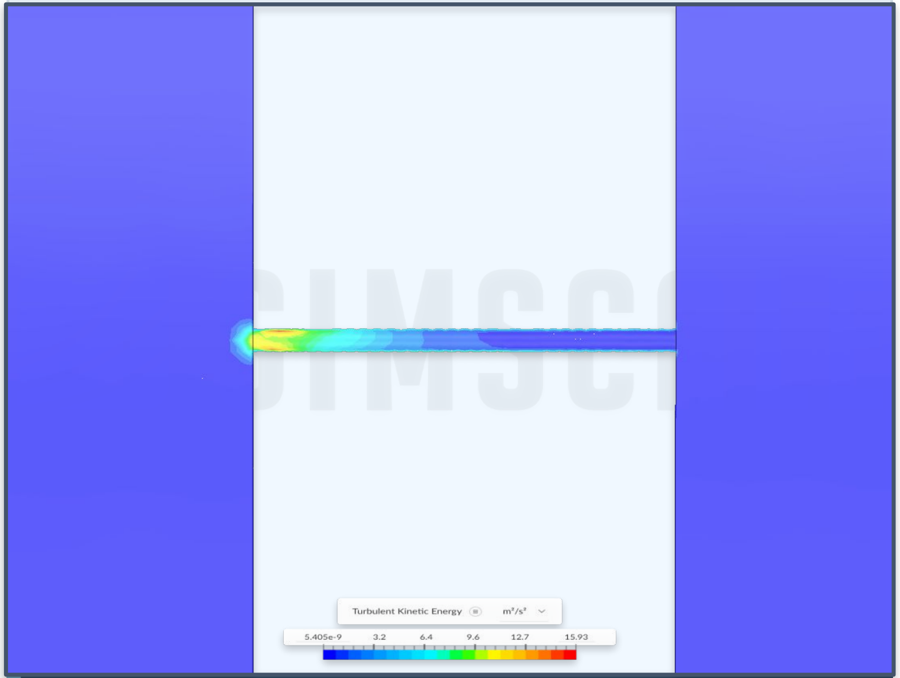

Fig. 6 — Turbulence Intensity / Kinetic Energy

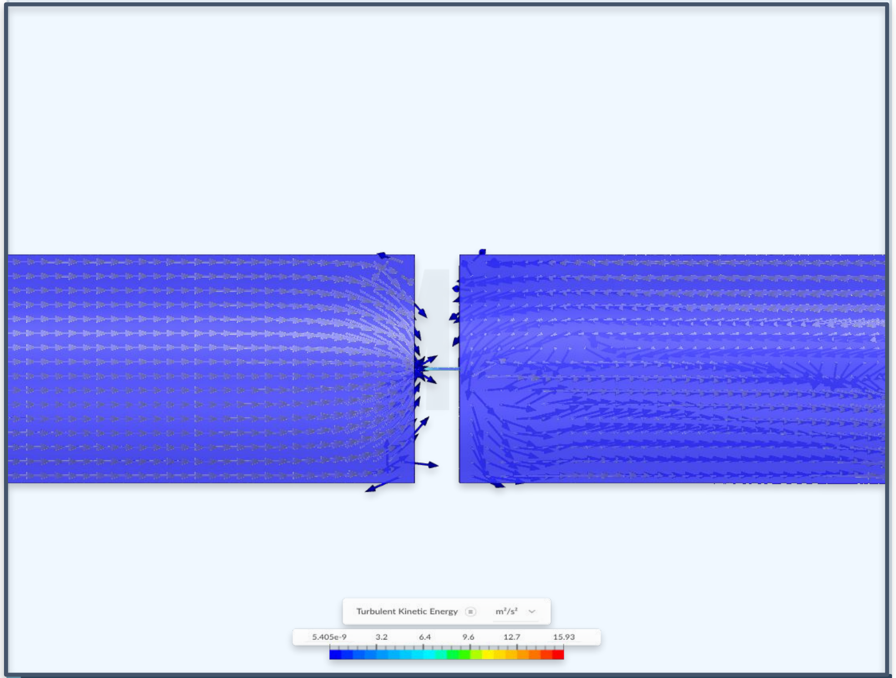

Fig. 7 — Velocity Vector Field

7 Calculations

Cutting planes: 1

Velocity Magnitude

Surface Area	3.81e-3 m^2
Maximum	25.36 m/s *
Average	1.346e-1 m/s *
Minimum	0 m/s *
Highlight in Model	
Integral	5.127e-4 (m/s)·m^2 *
Volumetric Flow Rate	3.53e-6 m^3/s *

Download regions as CSV*

Statistics are based on interpolated data. Learn more

01 Mass Flow Rate

$\dot{m}$ = Volumetric flow rate × density

$\dot{m} = 3.53 * 10^{-6}\ m^3/s \times 971\ kg/m^3 =$ **0.00343 kg/s**

02 Convert to GPH

3.53e-6 m^3/s × 264.17 gal/m^3 × 3600 s/hr

= **3.36 GPH**

03 Energy Loss

$Q = \dot{m} \times Cp \times \Delta T = 0.00343 \times 4{,}187 \times (82°C - 60°C)$

Q = 315.8 W = **1,078 BTU/hr per valve**

E = 315.8 W × 8,760 hrs/yr = 2,766,408 Wh/yr

= **2,766 kWh/yr per valve**

8 Conclusions & Discussion

01 Key Finding

The CFD simulation confirmed that even a 0.3mm gap at the valve seat is enough to produce significant leakage flow. There was a pressure drop of 379,212 Pa (55 psi) across the leakage gap at a inlet velocity of 25.36 m/s. This shows that with a small opening, water will be shot across the valve like a jet due to the pressure drop.

02 Engineering Insight

A leakage rate of 0.00343 kg/s (≈ 3.36 GPH) of hot water flowing continuously is a meaningful and ongoing waste of energy. Giedt has multiple reheat zones, and if multiple have the same leaking issue, this could cause a significant waste of energy. With the reaction of the cooling system, this creates even more energy waste.

03 Validation

We calculated Reynolds number of 9,440 in the main pipe confirms turbulent flow. This matches what we saw in our simulation Velocity Vector Field. As predicted by the Reynolds number, the downstream flow showed clear turbulent characteristics including recirculation and a chaotic wake behind the valve.

04 Lessons Learned

This was an educative process of the systems we regularly work with, but do not normally visualize. Being able to see the pressure and velocity distributions through a leaking valve gave us a much deeper understanding of the inside the piping. With more time, we would two way valve geometry, and how leakage behavior changes with different gap sizes and locations along the valve.

Thank You

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CFD Software: SimScale | Link to Project: https://www.simscale.com/projects/ccredo/valve_leakage_eng_103_863858843/